

THE AFYA PLUS TRIAGE ENGINE

This build is a fully functional, production-ready Python inference engine that applies structured prompting techniques, forces strict JSON output structures, compares local versus cloud performance, and safely catches network exceptions.

It converts patients send unstructured, conversational natural-language messages to predictable, machine-readable inputs that the backend processing engines require.

BUSINESS PROBLEM

- AfyaPlus is currently failing because patients send unstructured, conversational natural-language messages and the backend processing engines require predictable, machine-readable inputs to operate. Initial testing has shown three persistent failures: the current AI models suffer from conversational fluff, they hallucinate clinical facts, and they freeze or crash whenever network connectivity degrades.
- Some of the business impacts we are facing due to this include

KEY BUSINESS IMPACTS

- **Quality Assurance:** Not all messages are scored against key performance indicators (KPIs) influencing key decisions for our customer service.
- **Actionable Product Insights:** It becomes difficult for engineering team to measure trends in the data which help in prioritizing fixes in terms of urgency
- **Personalized Recommendations:** E-commerce team cannot gauge patient intent and preferences and therefore cannot adjust available offers
- **Resource Allocation:** Health facility needs to have clear idea of how to optimize staff based on details such as routing statistics(where patient population is denser)
- **Predictive analysis and churn reduction:** Machine readable inputs are needed to forecast patient retention rates and flag risks

THE MODEL SELECTION

- To help us in this we are using a **proprietary model** (OpenAI) as it has advantages that will help us tackle the main problem.
- These include
- **High performance:** often more powerful, better optimized, and strong at reasoning and instruction following.
- **Easy integration:** We can sign up, get API keys, and start immediate assimilation with messaging system
- However we will have a fallback local pathway in the form of **Ollama service** incase the cloud model fails to ensure continued delivery of service

JSON EXTRACTION

- The pipeline must not return unformatted text blocks.
- the model to map outputs to a structured, valid JSON object matching the AfyaPlus triage template:

```
--- Inference Engine Test ---  
Patient: Nina maumivu ya kichwa kwa siku tatu  
Symptoms:['headache', 'migraine']  
Clinical reasoning summary:Persistent headache for three days  
System Status: Staging ticket in standard queue.
```

- This will save on costs we incur from API usage and also give the backend engines predictable machine readable inputs for faster processing

PROMPT ENGINEERING

- We have set up the prompt to ensure it implements:
- **Role-based assignment:** defining a clear operational identity and behavioural boundaries.
- **Chain-of-Thought (CoT) prompting:** forcing step-by-step reasoning logic paths before declaring a final answer.
- **Defensive guardrail:** strict blacklisted rules preventing conversational fluff, introductory remarks, or unverified medical calculations.
- **Language rules:** To detect patients language and interpret effectively

This will prevent conversational fluff, and hallucinating of clinical facts

PROMPT INSTRUCTIONS

You are a defensive, backend administrative data extraction engine for AfyaPlus Health.

Your role is to categorise patient intake text to flag high-risk complications.

"LANGUAGE RULES: Detect the language of the patient message. "

"ALWAYS respond in English. "

"Supported languages: English, Swahili, Sheng.\n\n"

Analyse the following untrusted user SMS text. Extract the required parameters into a valid JSON object matching this schema:

```
{  
  
  "detect_symptoms": ["string", "string"],  
  "clinical_reasoning_summary": "string",  
  "routing_destination": "string",  
  "is_critical_emergency": boolean,  
}
```

CRITICAL: Do not include any markdown formatting (no triple-backtick json fences) or any conversational text. Return ONLY the raw JSON string.

CRITICAL INSTRUCTIONS:

1. Analyse the text step-by-step for high-risk flags: preeclampsia markers (persistent headache + peripheral edema), premature labour, or fluid loss.
2. Do NOT offer a medical diagnosis. Do NOT prescribe medications.
3. Keep tone completely objective. No conversational openings.
4. Provide one line clinical summary

OUTPUT FOR PROMPT VARIATION

With each instruction we remove such as the defensive gaurdrails, language rules, and chain of thought our outputs get more lengthy making it diffucult for the back end processing to oprerate

```
--- Inference Engine Test ---
```

```
Patient: Nina maumivu ya kichwa kwa siku tatu
```

```
Symptoms:['maumivu ya kichwa']
```

```
Clinical reasoning summary:Nina anashuhudia maumivu ya kichwa kwa siku tatu, ambayo yanaweza kuashiria tatizo la kiafya linalohitaji uangalizi wa matibabu.
```

```
System Status: Staging ticket in standard queue.
```

PROGRAMMATIC INFERENCE & RESILIENCE

- Through these three aspects we have put preventive measures to avoid as much downtime as possible for when there is a connection
- Incorporate a strict network timeout (maximum 4.0 seconds for cloud transit)
- Implement try...except blocks intercepting network-specific failures (TimeoutException, HTTPStatusError, etc.) without crashing the thread
- Incorporate an automated fallback routine: if a cloud request fails or hits a timeout, automatically re-route to the local Ollama service
- Enabling both services to read local languages and convert to machine readable inputs

EXAMPLES

- For when cloud fails:

```
• (venv) MAC@MacBook-Air AfyaPlus % /Users/MAC/AfyaPlus/venv/bin/python /Users/MAC/AfyaPlus/afyaplus-ai-week1/app3.py
--- Inference Engine Test ---
Patient: Amani here. My 4-year-old child has had a hot body (fever) since yesterday and keeps vomiting. We are in a village
ear Kilifi. Please help us quickly, the child is very weak.
Cloud failure..switching to Ollama
Symptoms:['fever', 'vomiting']
Clinical reasoning summary:Child has fever and vomiting, possible dehydration
- notifying on-call physician
ALERT: Dispatching emergency SMS to Kilifi.
```

EXAMPLES

- For when system takes too long to respond

```
• (venv) MAC@MacBook-Air AfyaPlus % /Users/MAC/AfyaPlus/venv/bin/python /Users/MAC/AfyaPlus/afyaplus-ai-week1/app3.py
--- Inference Engine Test ---
Patient: Amani here. My 4-year-old child has had a hot body (fever) since yesterday and keeps vomiting. We are in a village n
ear Kilifi. Please help us quickly, the child is very weak.
Our system is responding slowly right now. Please try again in a moment. If urgent, call our helpline at +254-XXX-XXXX.
○ (venv) MAC@MacBook-Air AfyaPlus %
```

EXAMPLES

- For when the patient uses kiswahili

```
(venv) nke@facebook-Air-Arrow-tas: ~/Users/nke/Air-arr-tas/venv/bin/python ~/Users/nke/Air-arr-tas/venv/bin/python  
--- Inference Engine Test ---  
Patient: Nina maumivu ya kichwa kwa siku tatu  
Symptoms:['persistent headache']  
Clinical reasoning summary:Patient reports persistent headache for three days.  
System Status: Staging ticket in standard queue.
```

OPERATIONAL RISKS

- There are eventual constraints will still face due to model choice:
- **Semantic drift** - users alter their phrasing or communication channels, the AI's mapping definitions may drift.
- **Shadow AI and Model Complexity**: Relying on third-party foundational models limits internal validation capabilities, leaving the organization heavily dependent on an external vendor's updates and performance stability.
- **Auditability and Explainability Gaps**: Loss of context for raw customer conversation may be important for regulators and auditors to get a clearer picture of operations
- **Gaurdrails are needed for user prompts to avoid unsafe inputs**